Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The direction cosines $l,\ m$ and n of two lines are connected by the relations $l+m+n=0$ and $lm=0$, then the angle between the lines is

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) 0

Q.2

A line makes 45° with positive x -axis and makes equal angles with positive y, z axes, respectively. What is the sum of the three angles which the line makes with positive x, y and z axes?

(A) 180°

(B) 165°

(C) 150°

(D) 135°

Q.3

What are the direction cosines of a line which is equally inclined to the positive directions of the axes?

(A) $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

(B) $\left\langle -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

(C) $\left\langle -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

(D) $\left\langle \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\rangle$

Q.4

The equation of a plane containing the line of intersection of the planes $2x$ - y - $4 = 0$ and $y + 2z$ - $4 = 0$ and passing through the point $1, 1, 0$ is :

(A) $x+3y+z=4$

(B) $x-y-z=0$

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(C) $x - 3y - 2z = -2$

(D) $2x - z = 2$

Q.5

A line makes the same angle α with each of the x and y axes. If the angle θ, which it makes with the z -axis, is such that $\sin^2\theta = 2\sin^2\alpha$, then what is the value of α ?

(A) $\pi/4$

(B) $\pi/6$

(C) $\pi/3$

(D) $\pi/2$

Q.6

A line makes angles θ, ϕ and ψ with x, y, z axes respectively. Consider the following :

1. $\sin^2\theta + \sin^2\phi = \cos^2\psi$

2.$\cos^2\theta + \cos^2\phi = \sin^2\psi$

3.$\sin^2\theta + \cos^2\phi = \cos^2\psi$

Which of the above is/are correct?

(A) 1 only

(B) 2 only

(C) 3 only

(D) 2 and 3

Q.7

The direction cosines of two lines are related by $l + m + n = 0$ and $al^2 + bm^2 + cn^2 = 0$. The lines are parallel if

(A) $a + b + c = 0$

(B) $a^{-1} + b^{-1} + c^{-1} = 0$

(C) $a = b = c$

(D) None of these

Q.8

If θ is the acute angle between the diagonals of a cube, then which one of the following is correct?

(A) $\theta < 30^\circ$

(B) $\theta = 60^\circ$

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(C) $30^{\circ} < \theta < 60^{\circ}$

(D) $\theta > 60^{\circ}$

Q.9

A line makes angles θ, ϕ and ψ with $\mathrm{x}, \mathrm{y}, \mathrm{z}$ axes respectively. Consider the following

1.$\sin^2\theta + \sin^2\phi = \cos^2\psi$

2. $\cos^2\theta + \cos^2\phi = \sin^2\psi$

3.$\sin^2\theta + \cos^2\phi = \cos^2\psi$

Which of the above is/are correct?

(A) 1 only

(B) 2 only

(C) 3 only

(D) 2 and 3

Q.10

Consider the following relations among the angles α, β and γ made by a vector with the coordinate axes

1.$\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$

II. $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

Which of the above is/are correct?

(A) Only I

(B) Only II

(C) Both I and II

(D) Neither I nor II

Q.11

If the direction cosines of a line are $\left(\frac{1}{c}, \frac{1}{c}, \frac{1}{c}\right)$ then

(A) $0 < \mathrm{c} < 1$

(B) $\mathrm{c} > 2$

(C) $\mathrm{c} > 0$

(D) $\mathrm{c} = \pm\sqrt{3}$

Q.12

If a line makes an angle of $\pi/4$ with the postive directions of each of x -axis and y -axis, then the angle that the line makes with the positive dircetion of the z-axis is

(A) $\pi/4$

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(B) $\pi/2$

(C) $\pi/6$

(D) $\pi/3$

Q.13

A line makes the same angle θ, with each of the x and z axis. If the angle β, which it makes with y -axis, is such that $\sin^2\beta = 3\sin^2\theta$, then $\cos^2\theta$ equals

(A) $\frac{2}{3}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{1}{5}$

Q.14

A line AB in three-dimensional space makes angles 45° and 120° with the positive x -axis and the positive y -axis respectively. If AB makes an acute angle q with the positive $z-$
axis, then θ equals

(A) 45°

(B) 60°

(C) 75°

(D) 30°

Q.15

The distance between two points P and Q is d and the length of their projections of PQ on the $\mathrm{co}-$ordinate planes are $\mathrm{d}_1,\ \mathrm{d}_2,\ \mathrm{d}_3$. Then $\mathrm{d}_1^2+\mathrm{d}_2^2+\mathrm{d}_3^2=\mathrm{kd}^2$ where K is

(A) 1

(B) 5

(C) 3

(D) 2

Answers & Solutions

Q.1 Answer:

$\frac{\pi}{3}$

Solution:

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Given, $l + m + n = 0$ **...(i)**

and $lm = 0$

i.e. either $m = 0$ **or** $l = 0$

If $l = 0$ **, then putting in equation (i), we get** $m = -n$

∴ Direction ratios are $0, -n, n$ **i.e.** $0, -1, 1$

If $m = 0$ **, then putting in equation (i), we get** $l = -n$

∴ Direction ratios are $-n, 0, n$ **i.e.** $-1, 0, 1$

$\therefore \cos\theta = \frac{0\times -1 + -1\times 0 + 1\times 1}{\sqrt{0^2 + -1^2 + 1^2}\sqrt{-1^2 + 0^2 + 1^2}} = \frac{1}{2}$

$\Rightarrow \quad \theta = \frac{\pi}{3}$

Q.2 Answer:

165°

Solution:

We know that sum of square of direction cosines $= 1$ **i.e.** $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow \quad \cos^2 45^\circ + \cos^2\beta + \cos^2\beta = 1$

(As given $\alpha = 45^\circ$ **and** $\beta = \gamma$ **)**

$\Rightarrow \quad \frac{1}{2} + 2\cos^2\beta = 1$

$\Rightarrow \quad \cos^2\beta = \frac{1}{4}$

$\Rightarrow \quad \cos\beta = \pm\frac{1}{2}$**, Negative value is discarded,**

Since the line makes angle with positive axes. Hence, $\cos\beta = \frac{1}{2}$

$\Rightarrow \quad \cos\beta = \cos 60^\circ$

$\Rightarrow \quad \beta = 60^\circ$

$\therefore \quad$ **Required sum** $= \alpha + \beta + \gamma = 45^\circ + 60^\circ + 60^\circ$

$= 165^\circ$

Q.3 Answer:

$\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

Solution:

Let $\ell, \mathrm{m}, \mathrm{n}$ **are the dc's of a line that is inclined equally at** α **to the** $+$ **ve direction of axes. Now,** $\ell = \cos\alpha, \mathrm{m} = \cos\alpha, \mathrm{n} = \cos\alpha$

Also, $\ell^2 + \mathrm{m}^2 + \mathrm{n}^2 = 1$

$$3\cos^2\alpha = 1$$
$$\cos\alpha = \frac{1}{\sqrt{3}}$$

∴ dc's of the line are : $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

Q.4 Answer:

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$x - y - z = 0$

Solution:

The required plane is

$2x - y - 4 + \lambda y + 2z - 4 = 0$

it passes through $1, 1, 0$

$\Rightarrow 2 - 1 - 4 + \lambda 1 - 4 = 0$

$\Rightarrow -3 - 3\lambda = 0 \Rightarrow \lambda = -1$

$\Rightarrow x - y - z = 0$

Q.5 Answer:

$\pi/4$

Solution:

Since $l^2 + m^2 + n^2 = 1$

$\therefore \cos^2\alpha + \cos^2\alpha + \cos^2\theta = 1 \quad \ldots\ldots\ldots (1)$

$(\because$ **A line makes the same angle** α **with** x **and** y **-axes and** θ **with z-axis) Also,** $\sin^2\theta = 2\sin^2\alpha$

$\Rightarrow 1 - \cos^2\theta = 2\left(1 - \cos^2\alpha\right) \left(\because \sin^2 A + \cos^2 A = 1\right)$

$\Rightarrow \cos^2\theta = 2\cos^2\alpha - 1$**.....(2)**

- From equation (i) and (ii) $2\cos^2\alpha + 2\cos^2\alpha - 1 = 1$

$\Rightarrow 4\cos^2\alpha = 2 \Rightarrow \cos\alpha = \pm\frac{1}{\sqrt{2}} \Rightarrow \alpha = \frac{\pi}{4}, \frac{3\pi}{4}$

Q.6 Answer:

2 only

Solution:

If a line makes angle θ, ϕ **and** ψ **with** $\mathrm{x}, \mathrm{y}, \mathrm{z}$ **axes respectively, then** $\cos^2\theta + \cos^2\phi + \cos^2\psi = 1$

$\Rightarrow \cos^2\theta + \cos^2\phi = 1 - \cos^2\psi$

$\Rightarrow \cos^2\theta + \cos^2\phi = \sin^2\psi$

$\therefore$ **Statement**

(2) is correct.

Q.7 Answer:

$a^{-1} + b^{-1} + c^{-1} = 0$

Solution:

For $\mathrm{n} = -(l + m)$**, the second relation gives** $a\vec{a}^2 + bm^2 + c(l + m)^2 = 0$ **or** $(a + c)l^2 + 2clm + (b + c)$

$m^2 = 0.$

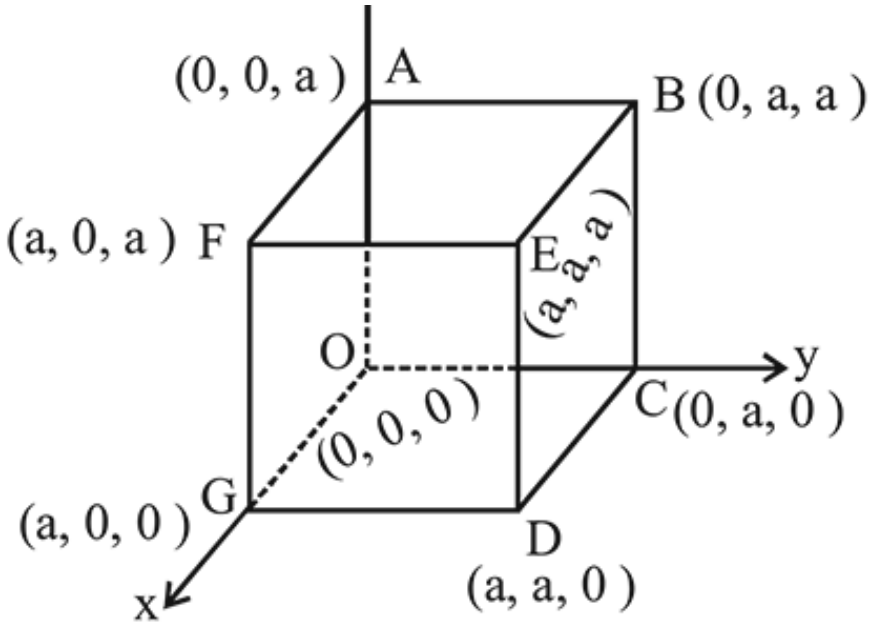
For parallel lines, the two roots must be equal $\Rightarrow 4c^2 - 4(b + c)(a + c) = 0 \Rightarrow ab + bc + ca = 0$

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Q.8 Answer:

$\theta > 60^\circ$

Solution:



Let there be cube of side 'a'. Co-ordinates of its vertices O, A, B, C, D, E, F have le marked in the figure. Diagonals are OE, FC, GB and AD. Direction ratios (dr_3) of these diagonals are :

$OE\langle(a-0),(a-0),(a-0)\rangle = (a, a, a)$

$FC\langle(-a, a, -a)\rangle; GB\langle(-a, a, a)\rangle$ **and** AD

$\langle(a, a, -a)\rangle$

Their des are :

OE, $\left\langle \frac{a}{\sqrt{a^2+a^2+a^2}}, \frac{a}{\sqrt{a^2+a^2+a^2}}, \frac{a}{\sqrt{a^2+a^2+a^2}} \right)$

$= \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

$AD, \left\langle \frac{a}{\sqrt{\Sigma a^2}}, \frac{a}{\sqrt{\Sigma a^2}}, \frac{-a}{\sqrt{\Sigma a^2}} \right\rangle = \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

Angle, θ**, between** AD **and** OE **is given by** $\cos\theta =$

$$\pm \frac{\frac{1}{\sqrt{3}}\times\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{3}}\times\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{3}}\times\frac{1}{\sqrt{3}}}{\sqrt{\left\{\left(\frac{1}{\sqrt{3}}\right)^2+\left(\frac{1}{\sqrt{3}}\right)^2+\left(\frac{1}{\sqrt{3}}\right)^2\right\}\left(\frac{1}{\sqrt{3}}\right)^2+\left(\frac{1}{\sqrt{3}}\right)^2+\left(-\frac{1}{\sqrt{3}}\right)^2}}$$

$= \frac{\frac{1}{3}}{1\times 1} = \pm\frac{1}{3}$

Since the cube is in positive octant, we take $+\frac{1}{3}$**.**

So, $\cos\theta = \frac{1}{3} \Rightarrow \theta > 60'$

[Since value of $\cos\theta$ **decreases as** θ **increases in 0 to** 90°**.** $\cos\theta = 1$ **when** $\theta = 0^\circ$ **and** $\cos\theta = 0$ **when**

$\theta = 90^\circ$ **]**

Q.9 Answer:

2 only

Solution:

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If a line makes angle θ, ϕ and ψ with x, y, z axes respectively, then $\cos^2\theta + \cos^2\phi + \cos^2\psi = 1$

$\Rightarrow \cos^2\theta + \cos^2\phi = 1 - \cos^2\psi$

$\Rightarrow \cos^2\theta + \cos^2\phi = \sin^2\psi$

$\therefore$ **Statement**

(2) is correct.

Q.10 Answer:

Only I

Solution:

We have, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow 2\cos^2\alpha + 2\cos^2\beta + 2\cos^2\gamma = 2$**...(i)**

$\Rightarrow 2\cos^2\alpha - 1 + 2\cos^2\beta - 1 + 2\cos^2\gamma - 1 = 2 - 3$

$\Rightarrow \cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$

Hence statement - I is correct. and now from (i), $1 - \sin^2\alpha + 1 - \sin^2\beta + 1 - \sin^2\gamma = 1$

$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Hence, only statement I is correct.

Q.11 Answer:

$c = \pm\sqrt{3}$

Solution:

Since d. c. of line are $\left\{\frac{1}{c}, \frac{1}{c}, \frac{1}{c}\right\}$

$\therefore \quad \frac{1}{c^2} + \frac{1}{c^2} + \frac{1}{c^2} = 1 \Rightarrow c^2 = 3 \Rightarrow c = \pm\sqrt{3}$

Q.12 Answer:

$\pi/2$

Solution:

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow \cos^2\gamma = 0$

$\Rightarrow \gamma = \pi/2$

Q.13 Answer:

$\frac{3}{5}$

Solution:

Let the direction cosines be $\cos\theta, \cos\theta, \cos\beta$**. Then**

$2\cos^2\theta + \cos^2\beta = 1 \quad \therefore \quad 2\cos^2\theta - 3\sin^2\theta = 0 \quad \Rightarrow \quad \cos^2\theta = \frac{3}{5}$

Q.14 Answer:

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60°

Solution:

$t = \cos 45^\circ = \frac{1}{\sqrt{2}}$

$\mathrm{m} = \cos 120^\circ = -\frac{1}{2}$

$n = \cos\theta$

where θ is the angle which line makes with positive z -axis. Now $\ell^2 + m^2 + n^2 = 1$

$\Rightarrow \frac{1}{2} + \frac{1}{4} + \cos^2\theta = 1$

$\cos^2\theta = \frac{1}{4}$

$\Rightarrow \cos\theta = \frac{1}{2}$

(θ Being acute)

$\Rightarrow \theta = \frac{\pi}{3}$

Q.15 Answer:

2

Solution:

Here, $d_1 = d\cos(90^\circ - \alpha)$

$d_2 = d\cos(90^\circ - \beta)$

$d_3 = d\cos(90^\circ - \gamma)$

$\Rightarrow d_1^2 + d_2^2 + d_3^2 = d^2\left(\sin^2\alpha + \sin^2\beta + \sin^2\gamma\right)$

$\Rightarrow d_1^2 + d_2^2 + d_3^2 = 2d^2; \quad \therefore k = 2$

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